

PATENT APPLICATION

Title:

A Network Configuration Adaptive Deployment System and Method Based on Physical Perception and Finite Element Domain Decomposition

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FIELD OF TECHNOLOGY

This invention relates to network automation, distributed systems, and cyber-physical control.

In particular, the invention relates to a computer-implemented method and system for adaptive deployment of network configuration, wherein deployment concurrency, execution sequencing, and scheduling behavior are dynamically adjusted based on real-time physical state metrics collected from network devices.

BACKGROUND

Modern network infrastructure deployments, including data centers, telecom networks, and cloud environments, involve configuration of large numbers of devices. Existing deployment systems typically use fixed concurrency limits or static execution sequences.

These static deployment mechanisms fail to account for real physical device conditions such as CPU load, hardware temperature, and device aging. As a result, deployment operations may overload devices, cause instability, or produce unpredictable convergence behavior.

Furthermore, existing deployment systems lack deterministic convergence control and cannot dynamically adjust deployment speed based on real-time physical constraints.

Accordingly, there is a need for a deployment system capable of adaptive concurrency control and deterministic convergence based on real-time physical state metrics.

SUMMARY OF THE INVENTION

The invention provides a system and method for adaptive network configuration deployment using physical state-based damping control and topology domain decomposition.

The system dynamically adjusts deployment concurrency and scheduling delay using a damping factor calculated from physical device metrics.

In one embodiment, the system calculates a damping factor K using:

$$K = 1 + \alpha(T - T_{\text{threshold}})/T_{\text{threshold}} + \beta L + \gamma A$$

where:

T represents device temperature

L represents device CPU load

A represents hardware aging factor

α, β, γ are weighting coefficients

The deployment concurrency window is controlled using:

$$\text{Concurrency_Window} = \text{Base_Window} / K$$

Deployment delay between steps is controlled using:

$$\text{Step_Delay} = \text{Base_Delay} \times K$$

The system monitors convergence using a residual defined as:

$$\text{Residual} = \|\text{State_actual} - \text{State_target}\|$$

where:

State_actual represents the current operational state of network devices

State_target represents the intended configuration state

Convergence is achieved when:

$$\text{Residual} < \varepsilon$$

where ε is a predefined convergence threshold.

The network topology is decomposed into subdomains, allowing parallel deployment within subdomains while maintaining global convergence stability.

BRIEF DESCRIPTION OF DRAWINGS

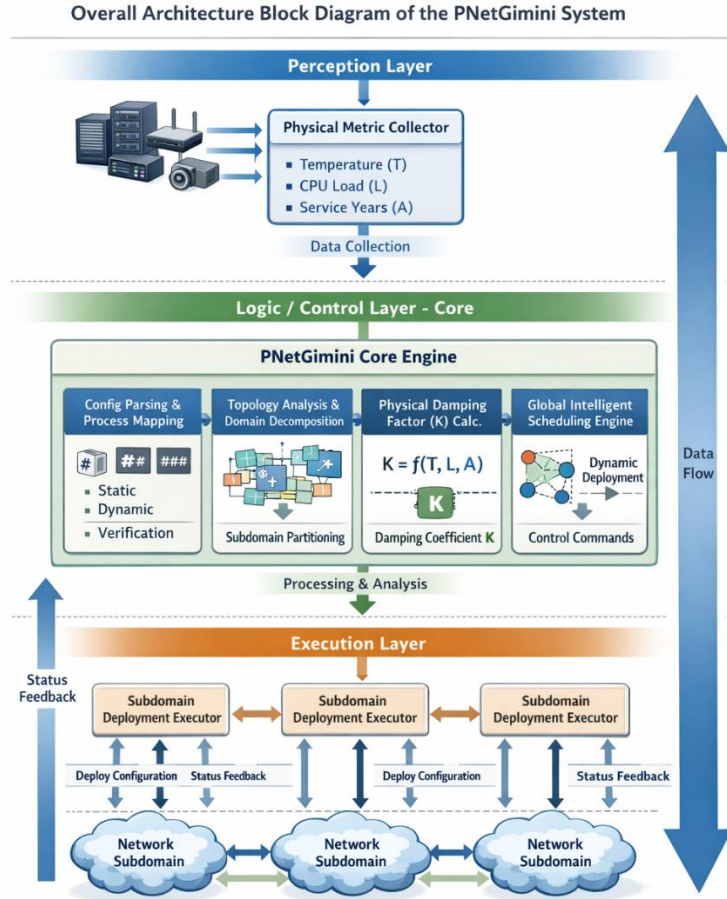


Figure 1 Overall Architecture of the PNetGimini System.

As shown in the figure, the system adopts a three-layer architecture design:

1. **Perception Layer (Upper):** Real-time physical state data such as temperature (T), CPU load (L), and service years (A) of network devices are acquired through a Physical Metrics Collector.
2. **Logic Layer (Middle, PNetGimini Core Engine):** Receiving data from the Perception Layer and the raw configuration, it processes them sequentially through modules including Configuration Parsing and Process Step Mapping, Topology Analysis and Domain Decomposition, and Physical Damping Factor Calculation. Finally, a Global Scheduler generates the dynamic deployment strategy.
3. **Execution Layer (Lower):** Multiple Subdomain Deployment Executors work in parallel. Based on scheduling instructions, they perform configuration deployment and state management on their respectively assigned network subdomains, and feed back the execution status to the Perception Layer, forming a closed-loop control.

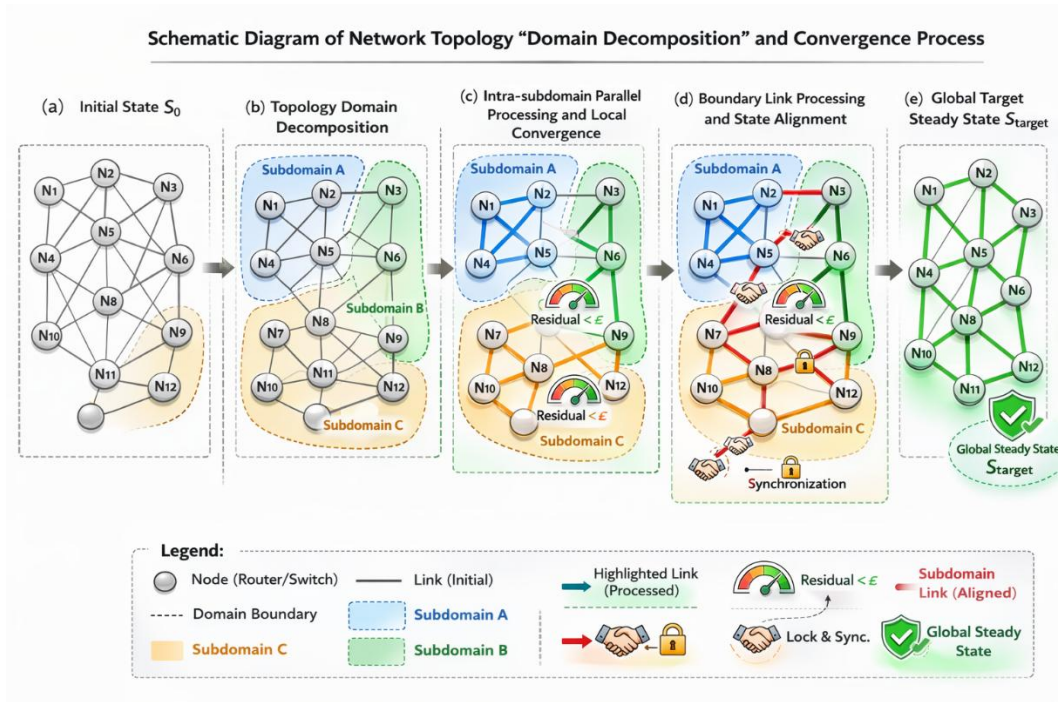


Figure 2 Topology Domain Decomposition and Convergence Process.

This figure illustrates the core “divide-and-conquer with coordinated convergence” strategy of the present invention, demonstrating a deterministic transformation mechanism from physical topology modeling to ordered logical execution.

- **(a) Initial State S_0 :**
The original network topology prior to configuration, representing a globally coupled system awaiting coordinated deployment.
- **(b) Domain Decomposition:**
This stage implements the technical means of regional decomposition. A complex, globally interdependent topology is partitioned into multiple non-overlapping subdomains. Each subdomain exhibits strong internal coupling and weak external coupling. By minimizing cross-domain dependencies, the invention transforms a monolithic global convergence problem into multiple parallelizable sub-problems, thereby establishing the structural foundation for concurrent processing.
- **(c) Intra-Subdomain Convergence:**
Within each subdomain, configuration tasks are executed independently and in parallel. Protocol states converge locally, achieving a stable intra-domain steady state. The residual within each subdomain satisfies the convergence threshold condition ($\text{Residual} < \epsilon$), indicating local equilibrium.
- **(d) Boundary Alignment:**
After local convergence is achieved, boundary links between subdomains are processed in a coordinated and deterministic manner. This controlled boundary synchronization integrates the locally converged states into a globally consistent system state. By sequencing inter-domain coordination after intra-domain stabilization, the invention effectively mitigates control plane oscillations and instability risks commonly associated with traditional large-scale concurrent deployments.
- **(e) Global Target Steady State S_{target} :**
The entire network reaches the intended final steady state, in which all nodes and links operate in a globally synchronized and stable configuration.

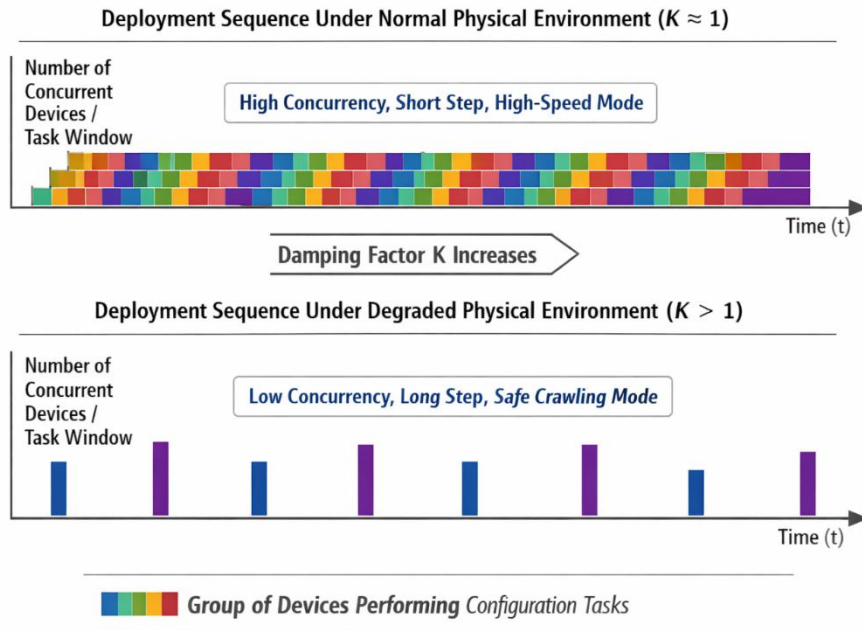


Figure 3 Damping Factor Effect on Deployment Concurrency and Timing.

Through the upper and lower comparative graphs, the control effect of the physical-field damping coefficient is intuitively illustrated:

- Upper Graph (Normal Mode, $K \approx 1$):
The deployment sequence is densely scheduled with a high degree of concurrency, thereby achieving optimal execution efficiency.
- Lower Graph (Degraded Mode, $K > 1$):
The deployment sequence becomes more sparsely distributed, and the concurrency window is reduced. The system automatically transitions into a “safe crawling mode,” sacrificing a portion of temporal efficiency in exchange for enhanced operational reliability.

DETAILED DESCRIPTION

In one embodiment, the adaptive deployment system described in this invention is referred to as the PNetGimini system (hereinafter referred to as "the system").

The system comprises:

- Physical Metric Collector Module
- Configuration Parsing Module
- Topology Domain Decomposition Module
- Damping Factor Calculation Module

- Global Deployment Scheduler
- Subdomain Deployment Executors

The Physical Metric Collector Module collects real-time device metrics including temperature, CPU utilization, and hardware health indicators.

The Damping Factor Calculation Module computes the damping factor K based on the collected physical metrics.

The Global Deployment Scheduler dynamically adjusts deployment concurrency and delay based on the damping factor.

The Topology Domain Decomposition Module partitions the network into subdomains to enable parallel deployment while maintaining convergence stability.

The system monitors convergence using the residual defined above and synchronizes deployment across subdomain boundaries.

When the residual falls below the convergence threshold ε , the deployment is considered complete.

The system prevents device overload by reducing concurrency when physical stress increases and improves deployment speed when physical conditions are favorable.

The system reduces manual intervention during large-scale network deployments by providing predictable convergence timelines.

The system improves operational safety, stability, and deployment efficiency.